Week 4: Limits, Continuity, and Intermediate Derivatives

4.1 Understanding Scenario — Chapter 4

The concepts covered: Second partial derivatives (pure and mixed) and the Hessian matrix.

Learning Outcome: At the end of this week you should be able to:

1. Understand the concept of second order partial derivatives and graphically.
2. Understand the concept of second order partial derivatives in matrix form.
3. Know the elements on the Hessian matrix and know how to find the pure and mixed second partial derivatives.
4. Understand the concept of second order partial derivatives in a multivariate setting.
5. Understand the concept of second order partial derivatives in a multivariate setting.

Reading: Chapter 4

Exercises: Chapter 4

1. The Challenge: From Total Change to Instantaneous Change

Why Instantaneous Rates Matter

In Week 1, we learned to describe motion using functions. Velocity tells us how fast an object is moving, and acceleration tells us how velocity changes. But real life doesn't move at a single constant speed. A car accelerates, a ball slows down, and a runner changes speed constantly. The question we need to answer is: **How do we describe a changing rate?**

Consider these everyday situations:

Context	Question	Why We Need
Medicine	How fast is the drug being absorbed?	Dosage timing and safety
Meteorology	How fast is the temperature rising or falling at this moment?	Predicting weather patterns
Finance	How fast is a stock price changing right now?	Trading decisions and risk management
Nature/Biology	How quickly do infections spread? How fast do ships and trucks move?	Public health and logistics

These questions cannot be answered by average rates. We need a tool that measures change at a single instant. This is the **instantaneous rate**, or the **derivative**.

From Average to Instantaneous: The Conceptual Leap

Suppose a car travels 100 km in 2 hours, covering a distance $\Delta x = 100$ km. The average velocity is:

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}} = \frac{\Delta x}{\Delta t} = \frac{100 \text{ km}}{2 \text{ hours}} = 50 \text{ km/h}$$

Average speed $= \frac{\text{Total distance}}{\text{Total time}} = \frac{\Delta x}{\Delta t} = \frac{100 \text{ km}}{2 \text{ hours}} = 50$ km/h

This tells us the total journey was 50 km/h on average. But it hides everything that actually happened. The car may have:

- Sat at a traffic light for 30 minutes
- Cruised at 80 km/h on the highway
- Stopped at a gas station
- Crawled through a school zone at 20 km/h

The average gives us a single number that describes none of these moments. To capture what happened at any moment, we need to shrink the time interval Δt until it approaches zero:

$$\text{Average speed} \to \text{Instantaneous speed} \quad \text{as} \quad \Delta t \to 0$$

Average speed $\to$ Instantaneous speed as $\Delta t \to 0$

1.1 Secant Lines and Tangent Lines

Consider a function $f(x)$ and two points on its graph: $(x_0, f(x_0))$ and $(x_1, f(x_1))$. The line connecting these points is called a secant line. Its slope is:

$$\text{Slope of secant line} = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

Slope of secant line $= \frac{f(x_1) - f(x_0)}{x_1 - x_0}$

As x_1 moves closer to x_0, the secant line rotates. When x_1 reaches x_0, the two points merge and the secant line becomes a tangent line. The slope of the tangent line is the instantaneous rate of change.

Figure 1: The Secant Tangent

1.2 Why Limits Matter

Consider the expression $\frac{0}{0}$. It looks undefined. But as $x \to 0$, $\frac{x}{x} = 1$.

$$\frac{x}{x} = 1$$

$\frac{x}{x} = 1$

This is indeterminate: we cannot divide by zero. But what happens when we approach it?

$$\frac{x}{x} = 1$$
$$\frac{0.1}{0.1} = 1$$
$$\frac{0.01}{0.01} = 1$$
$$\frac{0.001}{0.001} = 1$$
$$\frac{0.0001}{0.0001} = 1$$

The values approach 0 but both stay 1. This is the limit.

1.3 Evaluating Limits Algebraically

Strategy 1: Factoring [Example]

$$\lim_{x \to 2} \frac{x^2 - 4}{x - 2} = \lim_{x \to 2} \frac{(x-2)(x+2)}{x - 2} = \lim_{x \to 2} (x + 2) = 4$$

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Example 2: Evaluate $\lim_{x \to 3} \frac{x^2 - 9}{x - 3}$

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Answer:

$$\lim_{x \to 3} \frac{x^2 - 9}{x - 3} = \lim_{x \to 3} \frac{(x-3)(x+3)}{x - 3} = \lim_{x \to 3} (x + 3) = 6$$

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Strategy 2: Rationalize [Example: Square Root]

$$\lim_{x \to 0} \frac{\sqrt{x+1} - 1}{x} = \lim_{x \to 0} \frac{(\sqrt{x+1} - 1)(\sqrt{x+1} + 1)}{x(\sqrt{x+1} + 1)} = \lim_{x \to 0} \frac{x}{x(\sqrt{x+1} + 1)} = \lim_{x \to 0} \frac{1}{\sqrt{x+1} + 1} = \frac{1}{2}$$

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Example 3: Evaluate $\lim_{x \to 0} \frac{\sqrt{x+4} - 2}{x}$

Answer:

$$\lim_{x \to 0} \frac{\sqrt{x+4} - 2}{x} = \lim_{x \to 0} \frac{(\sqrt{x+4} - 2)(\sqrt{x+4} + 2)}{x(\sqrt{x+4} + 2)} = \lim_{x \to 0} \frac{x}{x(\sqrt{x+4} + 2)} = \lim_{x \to 0} \frac{1}{\sqrt{x+4} + 2} = \frac{1}{4}$$

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1.4 Why Continuity Matters for Derivatives

Theorem (Sketch). Any function that is differentiable at a point is continuous at that point. Equivalently, if a function is not continuous at a point, it cannot be differentiable at that point.

Model Building. A social worker plots a discontinuous client satisfaction model. Interview 1 and month 2 interview 2 are connected by a jump. Is the model differentiable at month 2?

2. The Derivative: Measuring Instantaneous Change

2.1 From Secant to Tangent

The average rate of change of $f(x)$ between x and $x + h$ is the slope of the secant line:

$$\text{Average rate of change} = \frac{f(x+h) - f(x)}{h}$$

Average rate of change $= \frac{f(x+h) - f(x)}{h}$

As $h \to 0$, the secant line approaches the tangent line at x. The limit of this ratio is the instantaneous rate of change, or the derivative:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

Figure 2: Secant Lines Approaching Tangent

2.2 Computing Derivatives from First Principles

Example 1: Find the derivative of $f(x) = x^2$.

Answer:

$$f'(x) = \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - x^2}{h} = \lim_{h \to 0} \frac{2xh + h^2}{h} = \lim_{h \to 0} (2x + h) = 2x$$

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Expand:

$$f'(x) = \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - x^2}{h} = \lim_{h \to 0} \frac{2xh + h^2}{h} = \lim_{h \to 0} (2x + h) = 2x$$

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Factor out h:

$$f'(x) = \lim_{h \to 0} \frac{2xh + h^2}{h} = \lim_{h \to 0} (2x + h) = 2x$$

$f'(x) = \lim_{h \to 0} \frac{2xh + h^2}{h} = \lim_{h \to 0} (2x + h) = 2x$

Therefore, $f'(x) = 2x$.

Interpretation: The slope of $f(x) = x^2$ the instantaneous rate of change at x is $2x$. At $x = 3$, the slope is 6; at $x = 10$, the slope is 20.

2.3 Basic Differentiation Rules

Computing derivatives from first principles is tedious. Fortunately, mathematicians have proved rules that work for all functions.

2.3 The Power Rule

$$\frac{d}{dx} x^n = n x^{n-1}$$

$\frac{d}{dx} x^n = n x^{n-1}$

This works for any real number n, including negatives and fractions:

Function	Derivative	Explanation
x^3	$3x^2$	Power rule: $3 \cdot x^{3-1} = 3x^2$
x^{-1}	$-x^{-2}$	Power rule: $-1 \cdot x^{-1-1} = -x^{-2}$
$x^{1/2}$	$\frac{1}{2}x^{-1/2}$	Power rule: $\frac{1}{2} \cdot x^{1/2-1} = \frac{1}{2}x^{-1/2}$
$x^{-1/2}$	$-\frac{1}{2}x^{-3/2}$	Power rule: $-\frac{1}{2} \cdot x^{-1/2-1} = -\frac{1}{2}x^{-3/2}$

Example 1: Differentiate $f(x) = x^5$.

Answer: $f'(x) = 5x^4$

Example 2: Differentiate $f(x) = x^{-2}$.

Answer: $f'(x) = -2x^{-3}$

Example 3: Differentiate $f(x) = x^{1/2}$.

Answer: $f'(x) = \frac{1}{2}x^{-1/2}$

2.3 The Constant Multiple Rule

$$\frac{d}{dx}[c \cdot f(x)] = c \cdot f'(x)$$

$\frac{d}{dx}[c \cdot f(x)] = c \cdot f'(x)$

You can "pull out" constant multiples.

Example 4: Differentiate $f(x) = 3x^4$.

Answer: $f'(x) = 3 \cdot 4x^3 = 12x^3$

2.3 The Sum/Difference Rule

$$\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$$

$\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$

Differentiate term by term.

Example 5: Differentiate $f(x) = x^3 - 5x^2 + 7x - 1$.

Answer:

$$f'(x) = 3x^2 - 5 \cdot 2x + 7 \cdot 1 - 0 = 3x^2 - 10x + 7$$

$f'(x) = 3x^2 - 5 \cdot 2x + 7 \cdot 1 - 0 = 3x^2 - 10x + 7$

Example 6: Differentiate $f(x) = x^2 - 3x^{-1} + 4x^{1/2}$.

Answer:

$$f'(x) = 2x^1 - 3 \cdot (-1)x^{-2} + 4 \cdot \frac{1}{2}x^{-1/2} = 2x + 3x^{-2} + 2x^{-1/2}$$

$f'(x) = 2x^1 - 3 \cdot (-1)x^{-2} + 4 \cdot \frac{1}{2}x^{-1/2} = 2x + 3x^{-2} + 2x^{-1/2}$

3. Geometric Interpretation: Tangent Lines

3.1 Equation of the Tangent Line

At point $(x_0, f(x_0))$, the tangent line has: Slope $= f'(x_0)$ Equation: $y - f(x_0) = f'(x_0)(x - x_0)$

Example 1: Find the tangent line at $x = 2$ for $f(x) = x^2$.

Answer: $f(2) = 4$, $f'(2) = 4$, so $y - 4 = 4(x - 2)$, or $y = 4x - 4$.

Example 2: Find the tangent line at $x = 1$ for $f(x) = x^3 - 2x + 1$.

Answer: $f(1) = 0$, $f'(1) = 1$, so $y - 0 = 1(x - 1)$, or $y = x - 1$.

3.2 Linear Approximation

The tangent line approximates $f(x)$ near x_0:

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0)$$

$f(x) \approx f(x_0) + f'(x_0)(x - x_0)$

This is a polynomial of degree 1 (a linear function).

Example: Approximate $\sqrt{4.1}$ using $f(x) = \sqrt{x}$ near $x_0 = 4$.

Answer: $f(4) = 2$, $f'(4) = \frac{1}{4}$, so $\sqrt{4.1} \approx 2 + \frac{1}{4}(0.1) = 2.025$. Actual value: $\sqrt{4.1} = 2.02485\ldots$

Error analysis: Approximation is off by 0.00015.

3.3 The Schedule Model: Setting Up the Optimization

Recall the Schedule problem from Week 3:

$$\text{Total Cost} = \text{Cost}(\text{Schedule}) + \text{Cost}(\text{Delay})$$

Total Cost $= \text{Cost}(\text{Schedule}) + \text{Cost}(\text{Delay})$

Cost(Schedule) is a linear function. Cost(Delay) is a quadratic function. We want to find the schedule that minimizes the total cost. This requires finding where the derivative equals zero.

4. Applying Instantaneous Rates to Science

4.1 Restricted Growth Rate

A biological population follows $G(x) = \frac{1000e^{0.5x}}{1 + 0.01(e^{0.5x} - 1)}$. Is this function differentiable?

Question: What is the instantaneous growth rate at $x = 10$ measured? **Analysis (qualitative):** Write an expression for the derivative. Hint: Use logic rules (Week 0). Or use approximation the difference between two nearby points (secant slope) approximates the derivative.

Interval	Average Rate (calculated)
$[10, 10.1]$	0.0000
$[10, 10.01]$	0.0000
$[10, 10.001]$	0.0000
$[10, 10.0001]$	0.0000

The limit appears to be approximately **0.0**. **(all calculations)** this is a horizontal tangent.

4.2 Radioactive Decay Rate

Iodine-131 decays according to $N(t) = N_0 e^{-0.086t}$. Half-life: 8 days.

Physical Meaning: The derivative $N'(t)$ tells us the **rate of decay** at time t. Radioactivity is proportional to the number of atoms. Fewer atoms, slower decay.

4.3 The Schedule Model: Setting Up the Optimization

Recall the Schedule problem from Week 3:

$$\text{Total Cost} = \text{Cost}(\text{Schedule}) + \text{Cost}(\text{Delay})$$

Total Cost $= \text{Cost}(\text{Schedule}) + \text{Cost}(\text{Delay})$

Cost(Schedule) is a linear function. Cost(Delay) is a quadratic function. We want to find the schedule that minimizes the total cost. This requires finding where the derivative equals zero.

5. Python: Numerical Differentiation and Visualization

5.1 Computing Derivatives Numerically

When we can't compute derivatives by hand, we approximate them with code. Here's the pattern for applying finite difference to arbitrary functions:

```
import numpy as np
import matplotlib.pyplot as plt

# Define any function
def f(x):
    return x**3 - 2*x + 1

# Compute derivative at point
x = 2.0
h = 0.001
f_prime = (f(x + h) - f(x - h)) / (2 * h)

print(f"Derivative at x = {x}: {f_prime:.6f}")
# Output: Derivative at x = 2.0: 10.000000
```

5.2 Visualizing Secant Lines Approaching Tangent

```
# Create data for plotting
x_vals = np.linspace(0, 4, 100)
y_vals = f(x_vals)

# Secant lines with different h values
h_values = [1.0, 0.5, 0.1, 0.01]

plt.figure(figure_text="Secant Lines Approaching Tangent")
plt.plot(x_vals, y_vals, 'b-', linewidth=2, label='f(x) = x³ - 2x + 1')

for h in h_values:
    slope = (f(x + h) - f(x - h)) / (2 * h)
    y_secant = f(x) + slope * (x_vals - x)
    plt.plot(x_vals, y_secant, '--', alpha=0.7, label=f'h = {h}')

plt.plot(x, f(x), 'ro', markersize=10)

plt.xlim(0, 4)
plt.ylim(-2, 12)
plt.grid(True, alpha=0.3)
plt.legend(loc='upper left')
plt.title('Secant Lines Approaching Tangent')
plt.xlabel('x')
plt.ylabel('f(x)')
plt.savefig('secant_lines.jpg', dpi=150, bbox_inches='tight')
plt.show()
```

5.3 Looking Ahead: Week 5

With the basic differentiation rules mastered, Week 5 will introduce:

- **Product rule:** $(fg)' = f'g + fg'$
- **Quotient rule:** $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$
- **Chain rule:** $(f(g(x)))' = f'(g(x)) \cdot g'(x)$
- **Implicit differentiation:** Finding derivatives when equations mix x and y
- **Optimization problems:** Finding maxima and minima using derivatives

These tools will allow you to analyze and model optimization problems like the profit maximization exercise (Q20) on next week.

Summary: Key Formulas for Week 4

Concept	Formula
Average rate of change	$\frac{f(x+h) - f(x)}{h}$
Derivative (definition)	$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$
Power rule	$\frac{d}{dx} x^n = n x^{n-1}$
Constant multiple	$\frac{d}{dx}[c \cdot f(x)] = c \cdot f'(x)$
Sum/difference	$\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$
Tangent line	$y - f(x_0) = f'(x_0)(x - x_0)$
Linear approximation	$f(x) \approx f(x_0) + f'(x_0)(x - x_0)$
Secant slope (numerical)	$\frac{f(x+h) - f(x-h)}{2h}$

Exam Alignment

Question	Topic	Week 4 Skill Required
Q17	Sequence	Recognize the sequence from data points
Q18	Sequence	Extrapolate the sequence
Q19	Sequence	Analyze the sequence
Q20	Optimization	Setting $f'(x) = 0$ to find MPS model-ept
Q21	Optimization	Differentiate profit function, solve for x

References

- Schaefer, M. (2026). Notes consolidation of quantitative decisions and economics in relation to the development of a model for the optimization of a schedule. Unpublished manuscript.
- Board of Canada. (n.d.). *MESH*.
- Stewart, J. (2016). *Calculus: Early Transcendentals* (8th edition). Cengage Learning.

Next Week: 5 – Differentiation Techniques and Optimization. Plan for reading Chapter 5.